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Computer Networks

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Lab 05

How to Connect Two LAN

Department Of Software Engineering

OBJECTIVE

The objective of this lab is to learn how to connect two Local Area Networks (LANs) using a switch and assign IP addresses to PCs to enable communication between them.

INTRODUCTION

A Local Area Network (LAN) is a group of connected devices within a limited area such as a home, office, or campus. To allow communication between two different LANs, a networking device such as a router/switch is required. This lab demonstrates how two LANs can be interconnected and configured using simulation software.

REQUIREMENTS

- 2 Switches
- 6 PCs
- Straight-through cables (for PC to switch)
- Dotted cable (for switch-to-switch connection)

NETWORK DESIGN:

- Two LANs are created using two switches.
- Each switch connects to 3 PCs.
- Switches are interconnected to allow communication between both LANs.

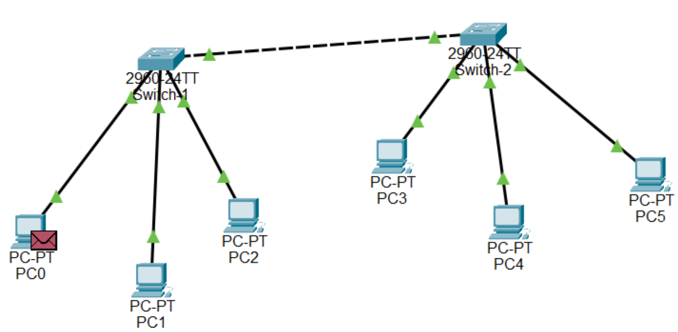

Figure 1:PCs and Switch Connection

◆ IP ADDRESSING:

- PC0 → 192.168.1.1
- PC1 → 192.168.1.2
- PC2 → 192.168.1.3
- PC3 → 192.168.1.4
- PC4 → 192.168.1.5
- PC5 → 192.168.1.6

◆ CONNECTIONS:

- PCs are connected to switches using straight cables (—).
- Switch1 is connected to Switch2 using a dotted cable (…..).

⚙️ WORKING EXPLANATION (DATA TRANSMISSION)

When a message is sent from **PC0 to PC4**, the process is:

- PC0 sends the request to **Switch 1**

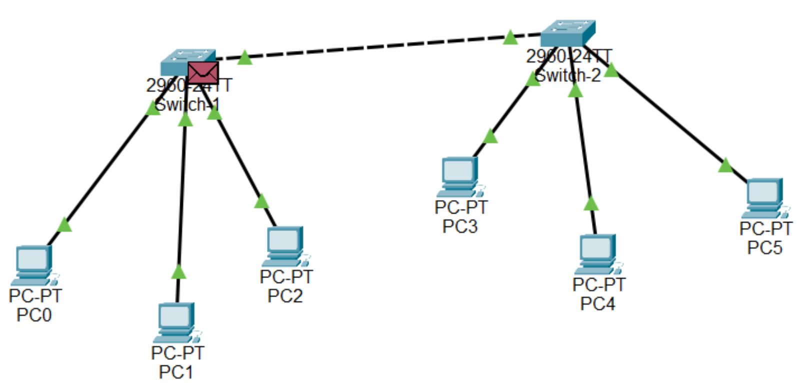

Figure 2 PC1 to switch 1

- Switch 1 forwards the request to **Switch 2**

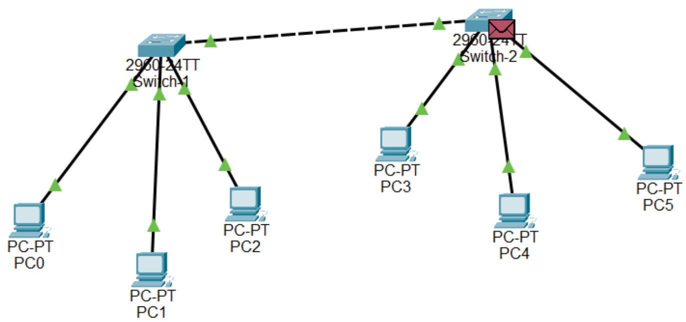

Figure 3 switch 1 to switch 2

- Switch 2 delivers the request to **PC4**

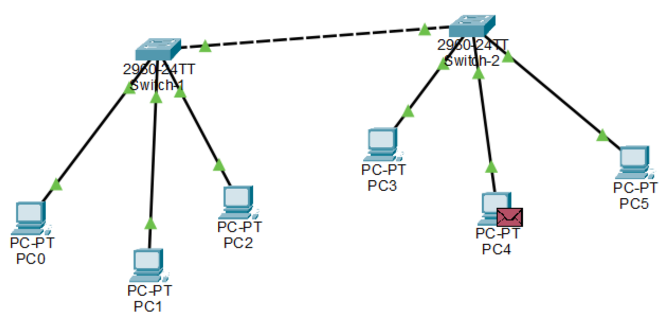

Figure 4 switch 2 to PC4

- PC4 sends a response back to **Switch 2**

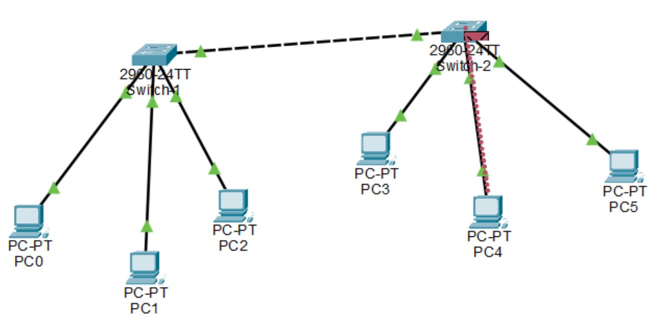

Figure 5 PC4 to switch 2

- Switch 2 forwards it to **Switch 1**

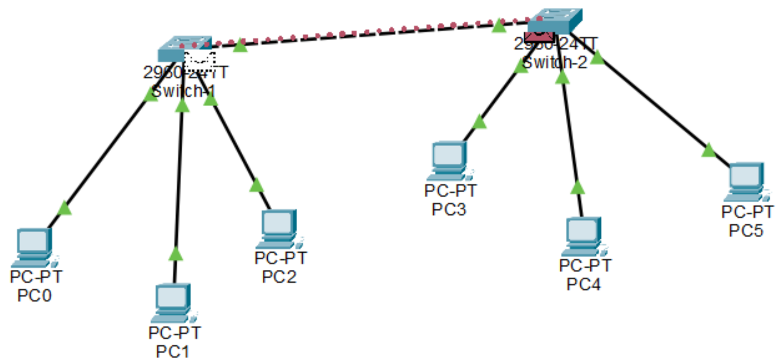

Figure 6 switch 2 to switch 1

- Switch 1 finally delivers it to **PC0**

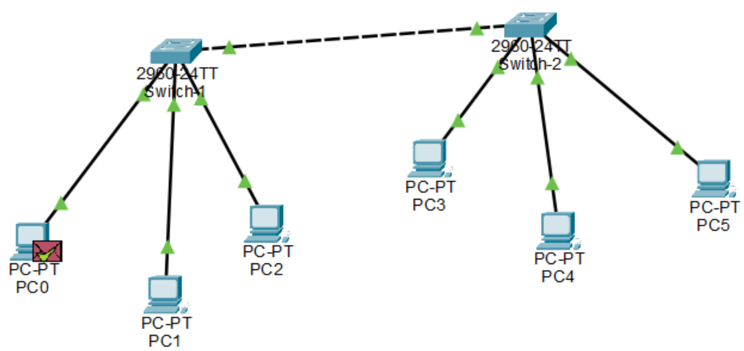

Figure 7 switch 1 to PC0

This confirms successful communication between the two LANs.

💻 SIMULATION PANEL

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch-1	ICMP
	0.002	Switch-1	Switch-2	ICMP
	0.003	Switch-2	PC4	ICMP
	0.004	PC4	Switch-2	ICMP
	0.005	Switch-2	Switch-1	ICMP
	0.006	Switch-1	PC0	ICMP

Figure 8 Simulation panel